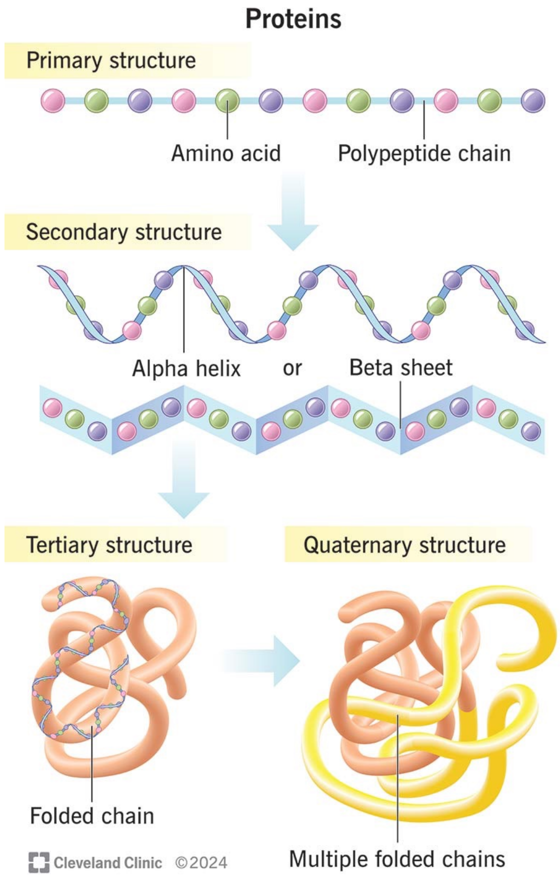
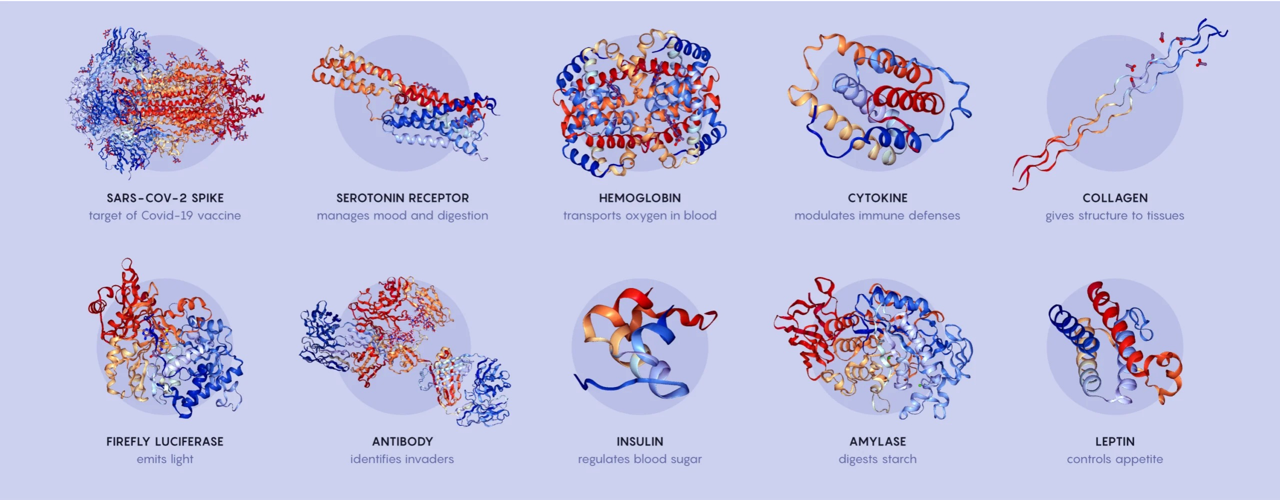
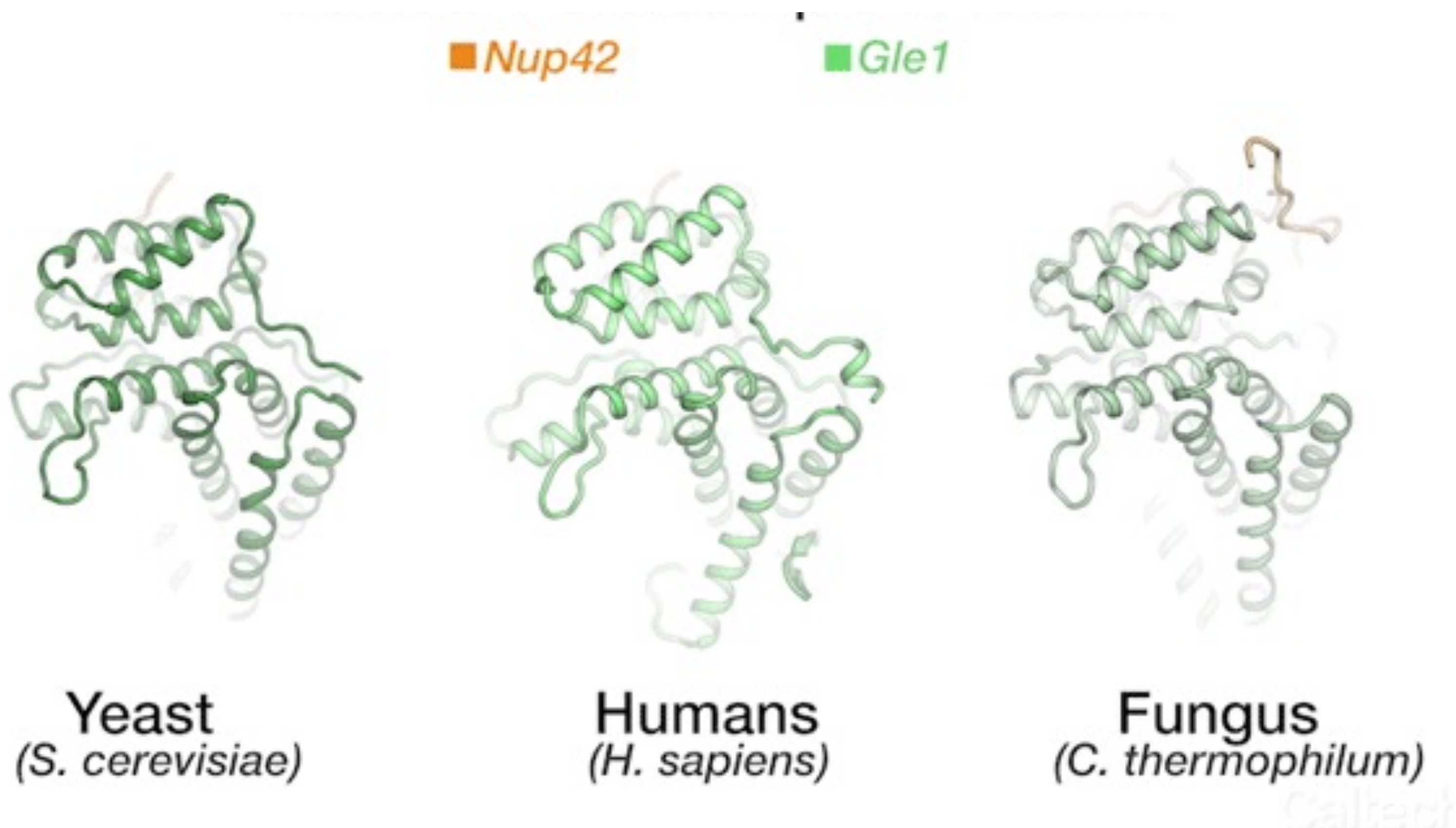
A brief introduction on Protein Structure Prediction and Protein Language Models

Amino acid sequences is **folded** to form a complex structure

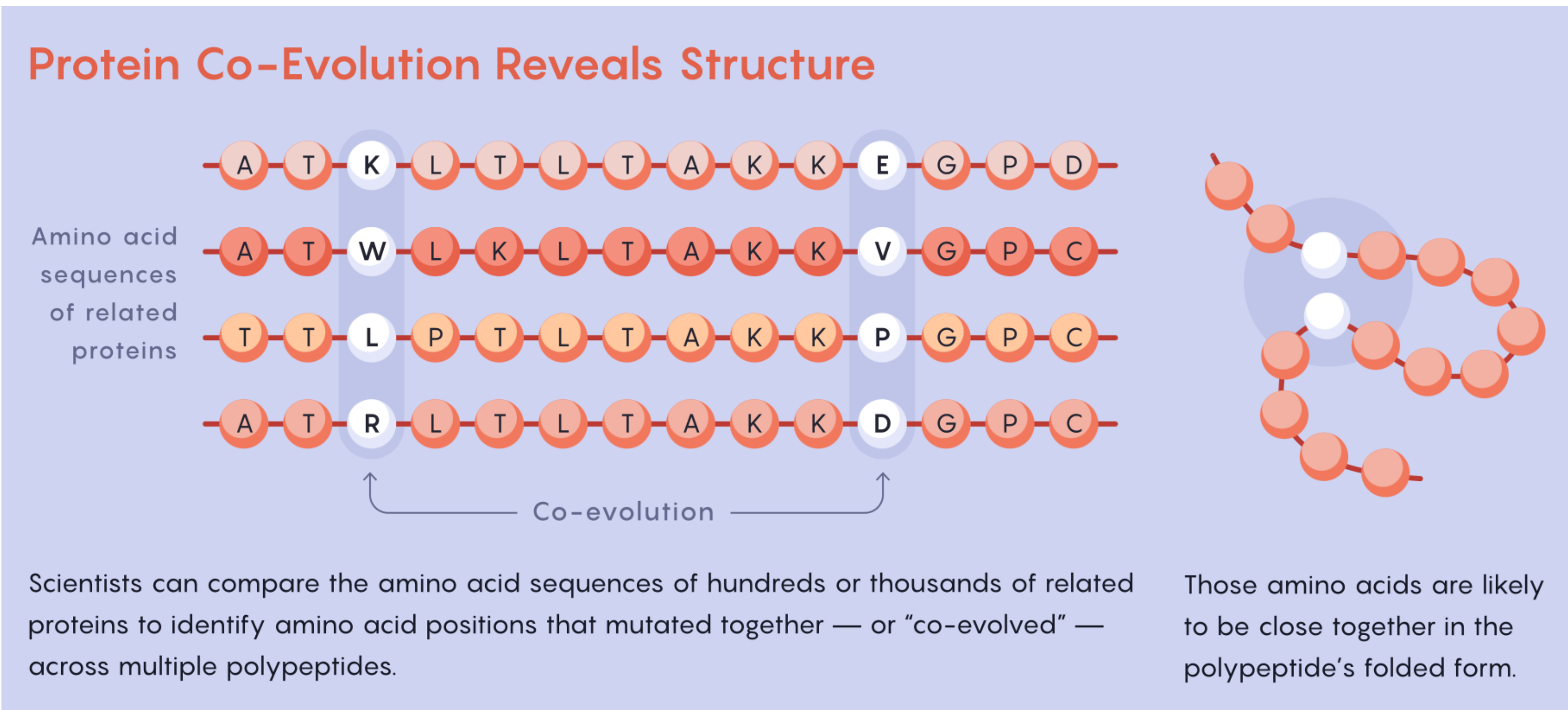




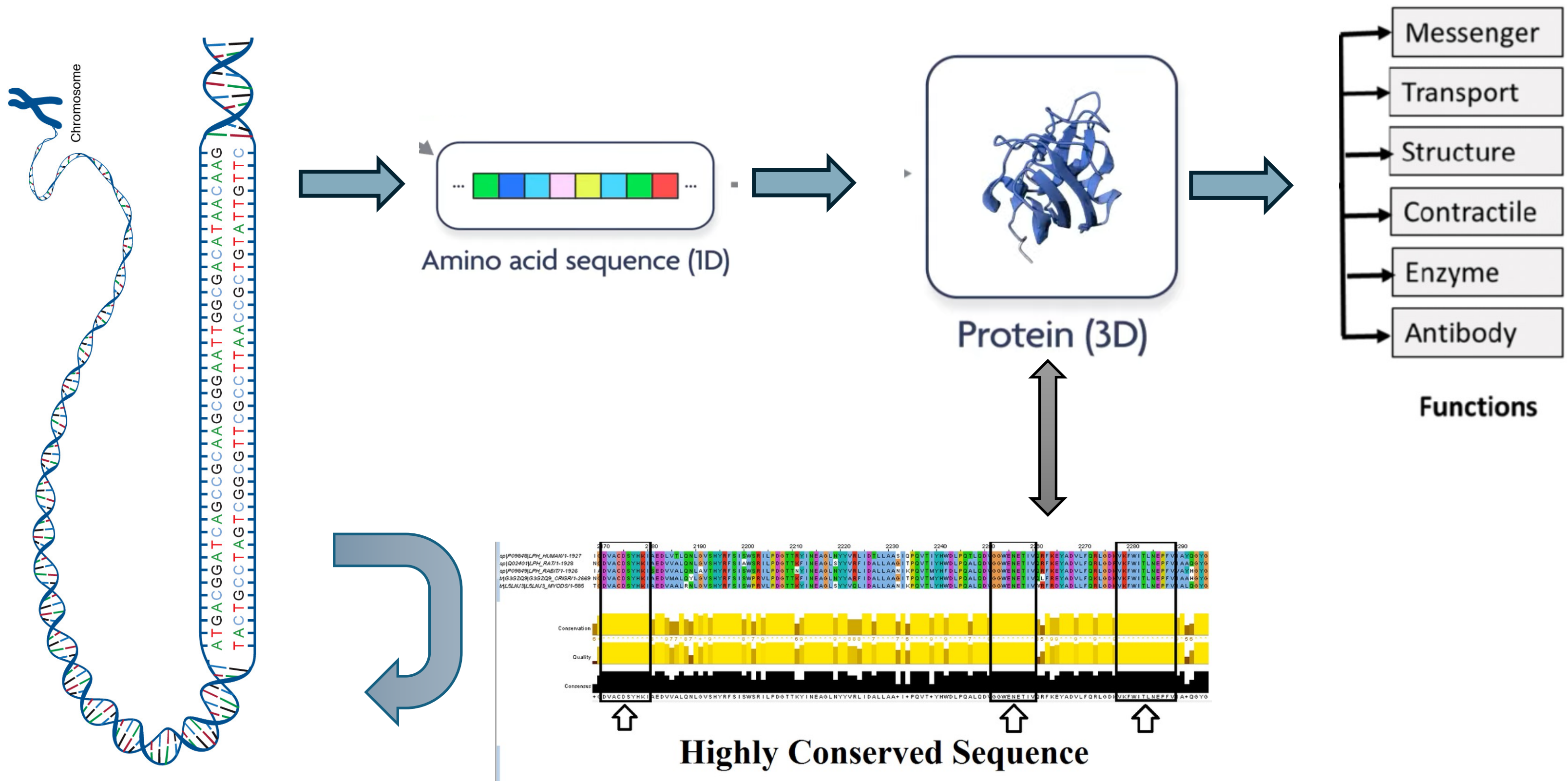
Nature reuses solutions and structures



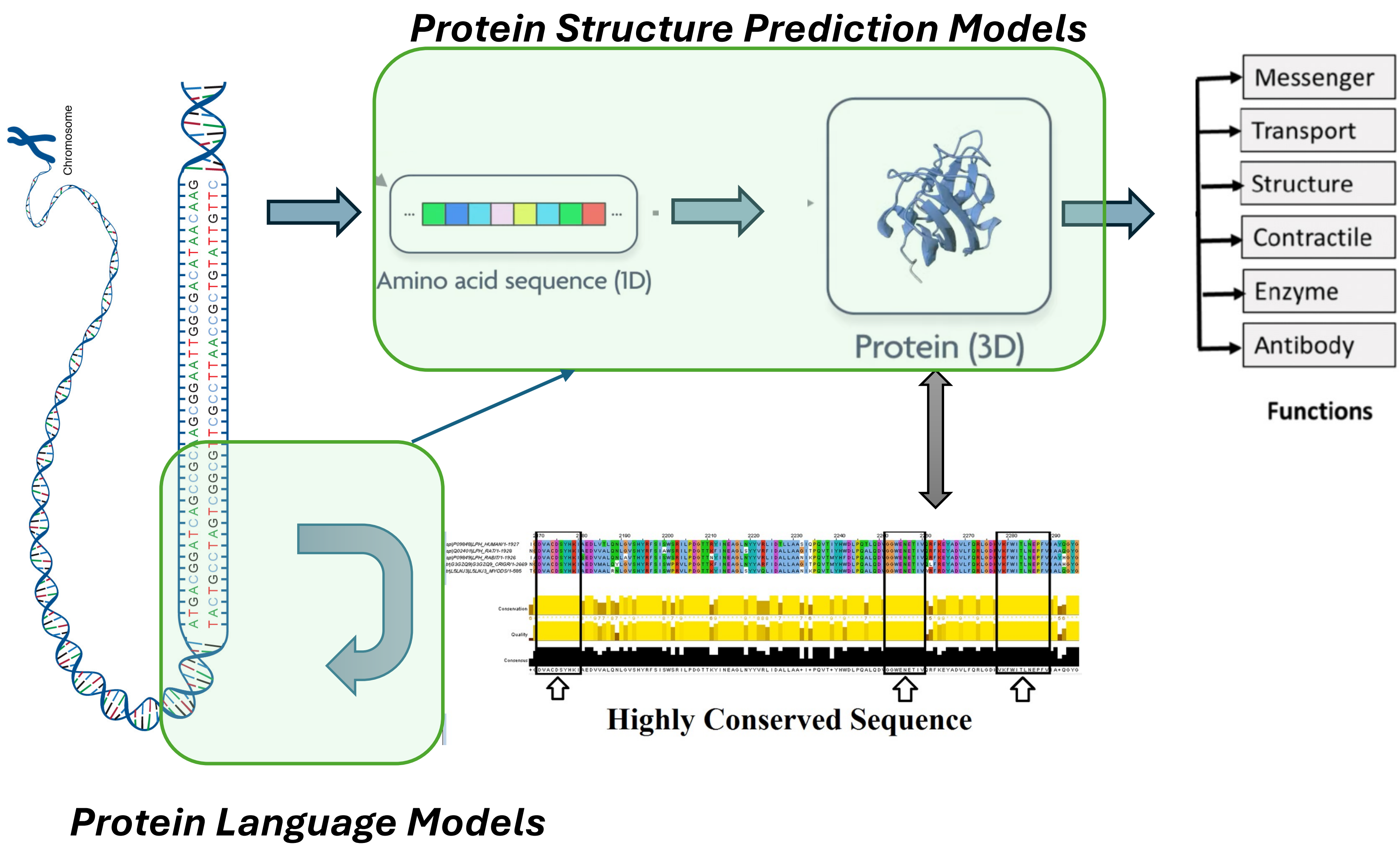
Co-Evolution leads to conserved structure



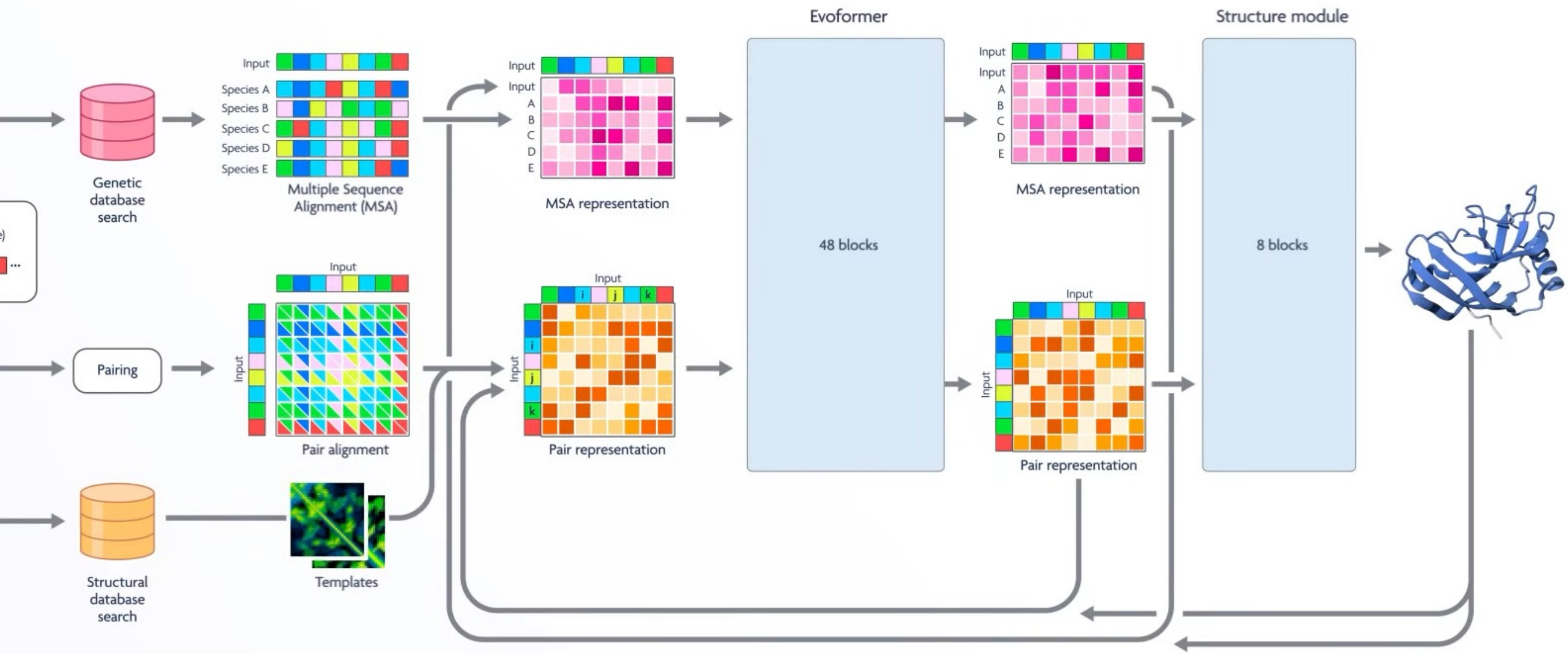
What is the big challenge ?



AI models in protein models : Two main category

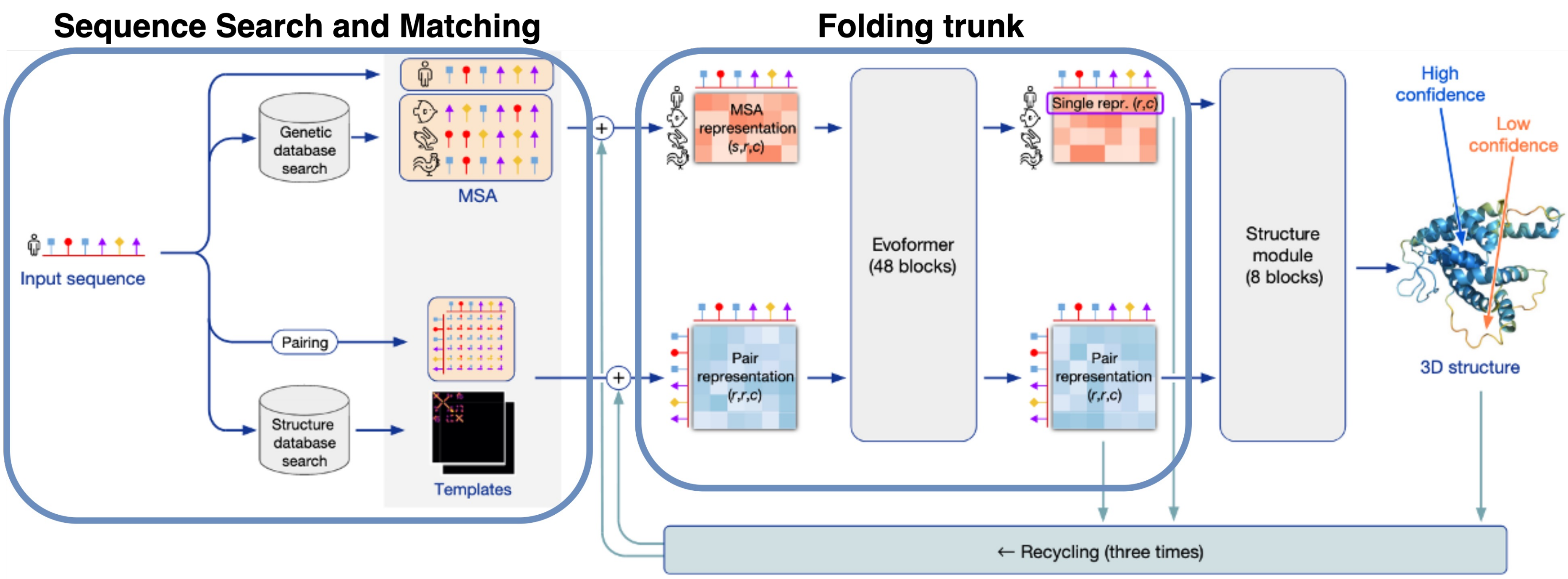


AlphaFold 2 Architecture



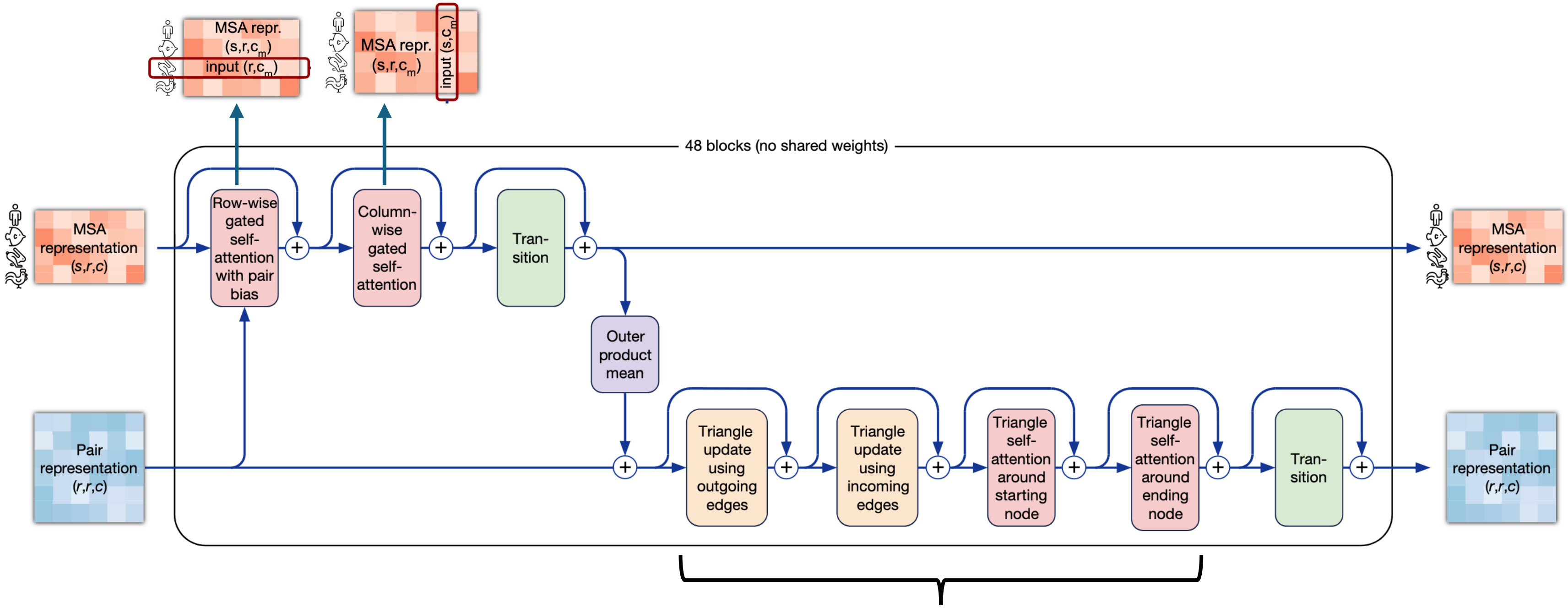
Source: NEJM

A closer look at AlphaFold2 architecture

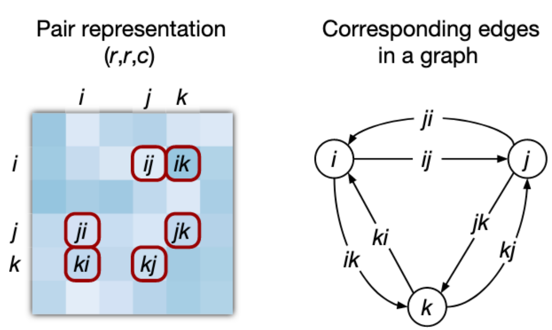


Jumper et al.

A closer look at folding trunk



Triangle attention applies the triangle inequality constraint

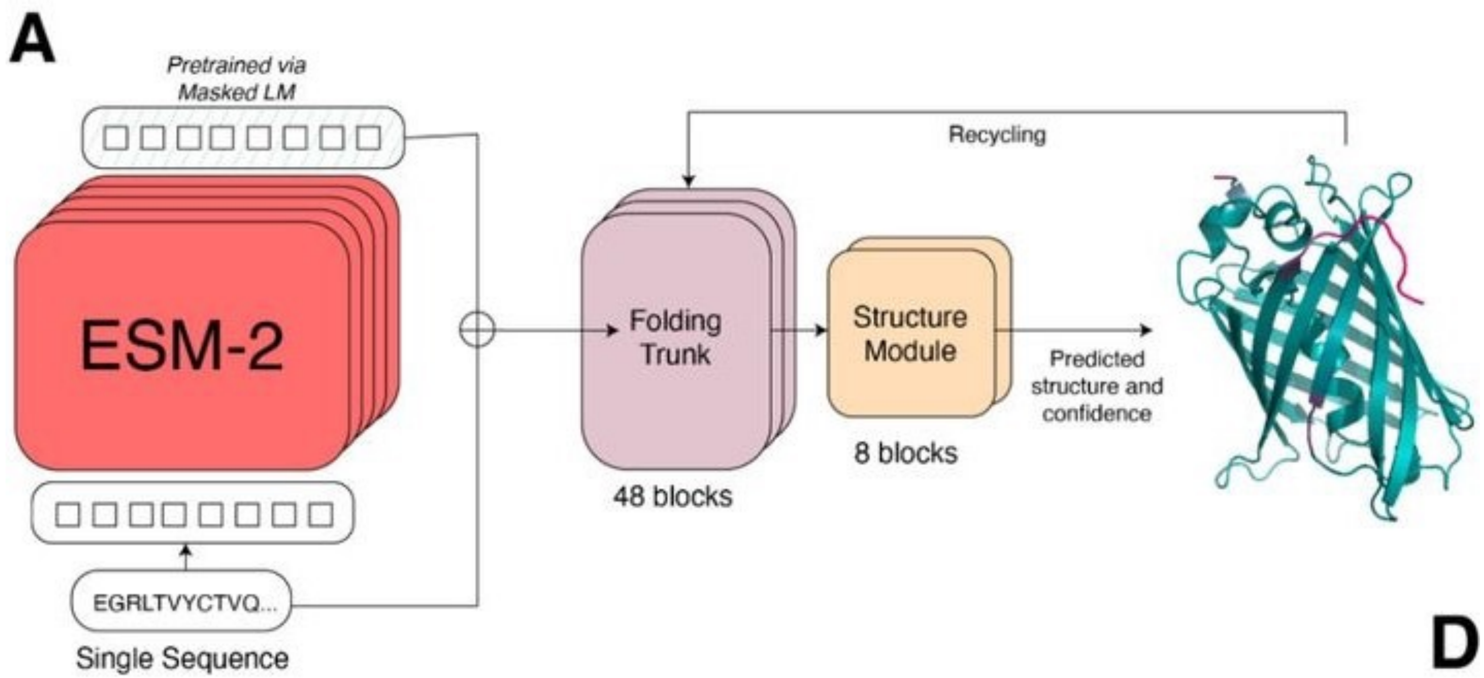


Triangle inequality theorem

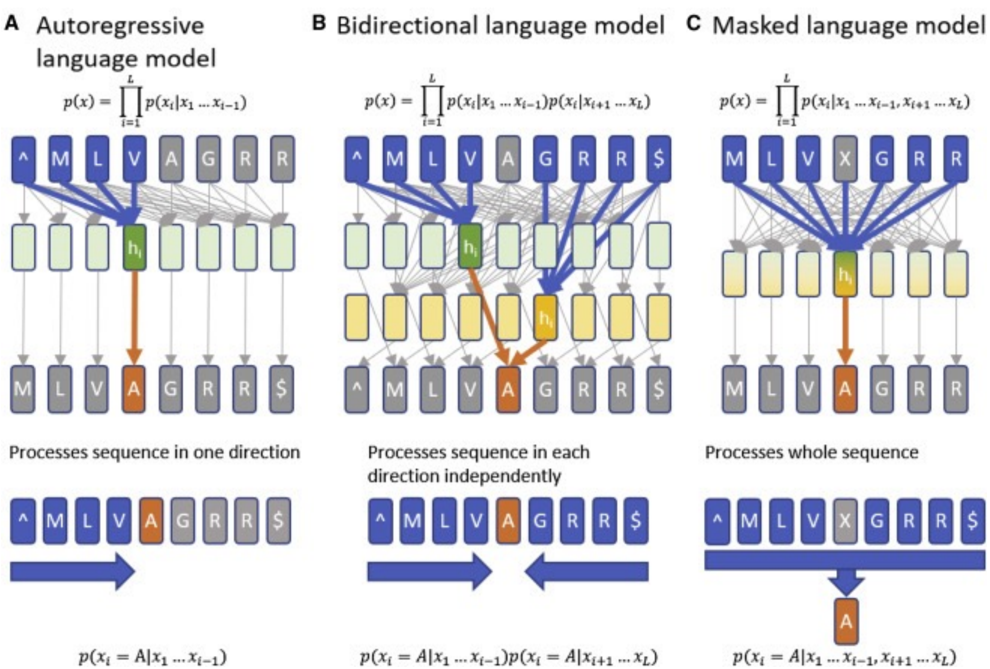
$$ij + ik \geq jk$$

Jumper et al.

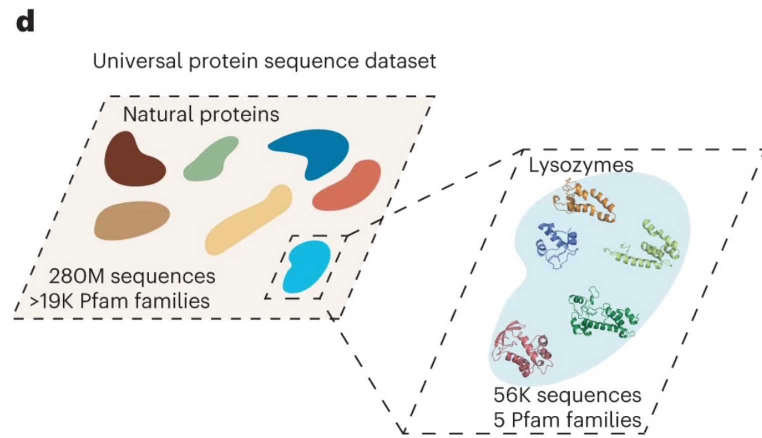
What about Protein Language Models ?



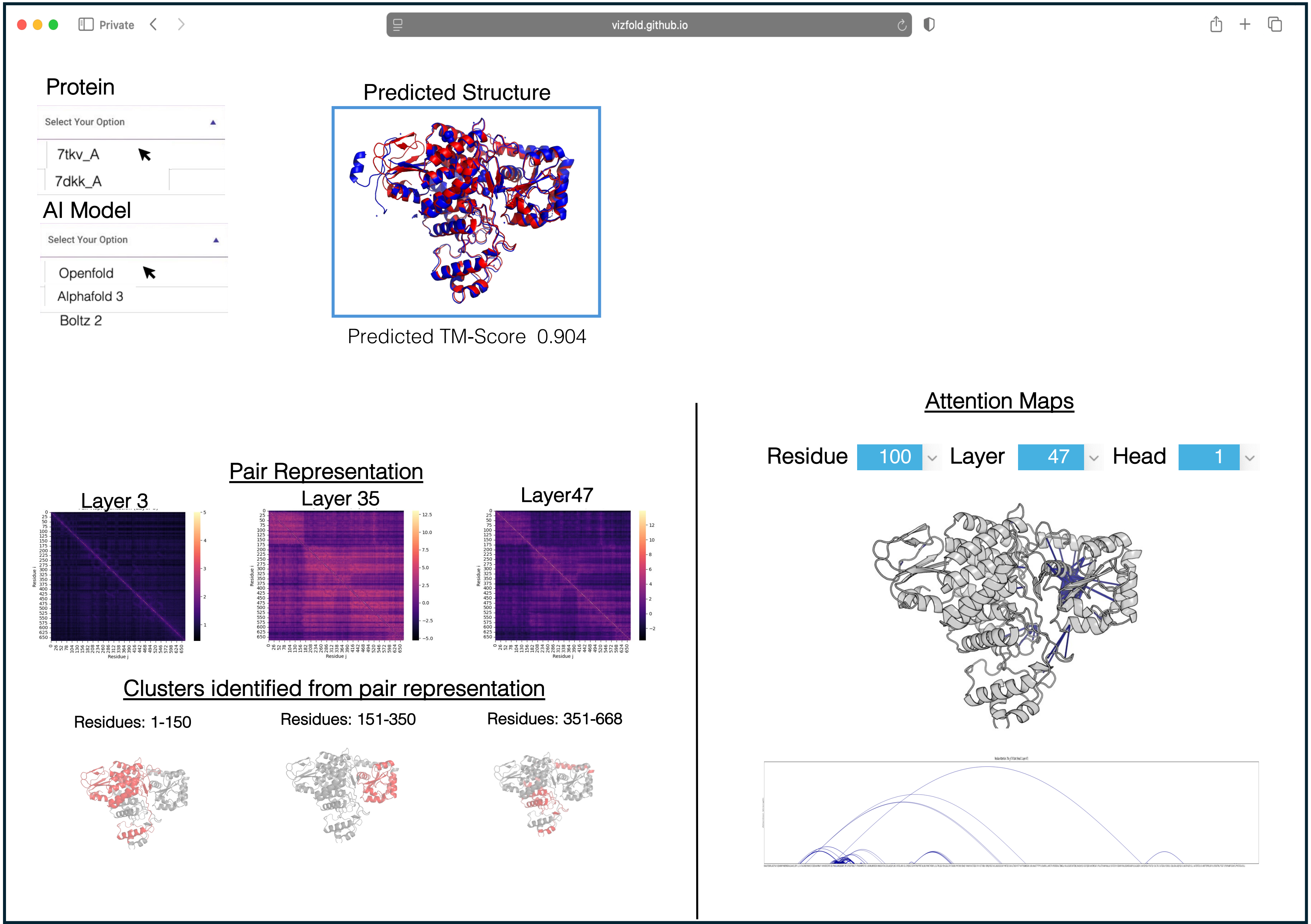
How are PLM trained ?



PLM embeddings have information about protein functions



Goal : Web Portal for Inference and Interpretability Trace for AI-based Protein Models



Slides by Devin Fromond

TA for Apache Airavata VIP Class

Refer: https://www.tamarind.bio/

Toward Modular and Interpretable Protein AI Pipelines

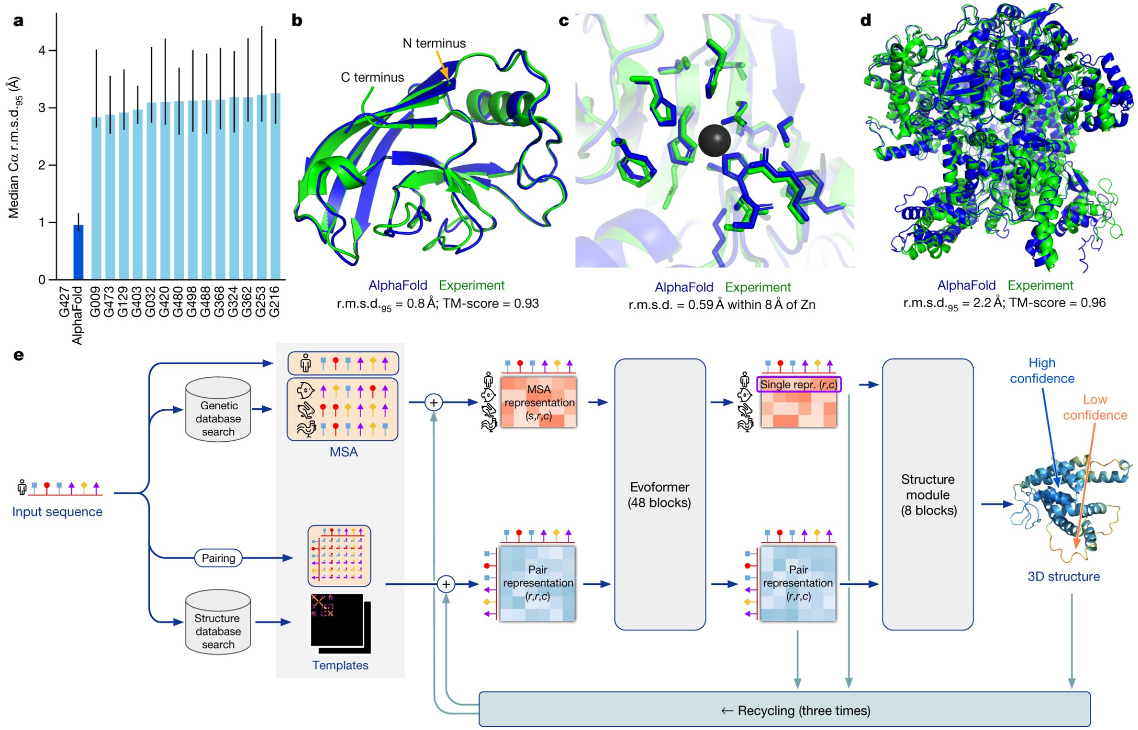
Apache Airavata | VIP VYL

Georgia Tech

Vizfold Overview

Instead of treating the model as a black box, VizFold tries to expose the intermediate representations inside the model, such as:

- embeddings
- attention maps
- intermediate geometric structures



What Could be more Interesting…

Current VizFold Pipeline

Protein Sequence → Run Full Model → Predicted 3D Structure

VizFold lets us inspect some intermediate signals, but the pipeline is mostly linear and fixed.

Future VizFold Pipeline

Protein Sequence → Feature Extraction → Input Embeddings → Attention / Evoformer Blocks → Pair & Single Representation → Structure Model → 3D Coordinate Prediction → Predicted Protein Structure

Feature Extraction:
- MSA
- Templates
- Sequence Features

Input Embeddings:
- Sequence
- Pair Representations

Attention / Evoformer Blocks:
- Refine residue relationship

Pair & Single Representation:
- Intermediate Structural Signals

Structure Model:
- Geometric Reasoning

Researchers could: inspect intermediate outputs, modify parameters, rerun downstream stages, compare results.

Two Directions that Enable This

Modularization Intermediate Layers

Expose pieces of the pipeline so they can be run independently:

```
run_embeddings()
run_attention()
run_structure()
```

This would allow intermediate visualization, partial pipeline execution, experimentation with parameters.

Standardizing Module Inputs/Outputs

Define a consistent interface between modules:

```
module(input_data, parameters) → output data
```

This makes it easier to plug in new tools, experiment with intermediate states, and test modifications.

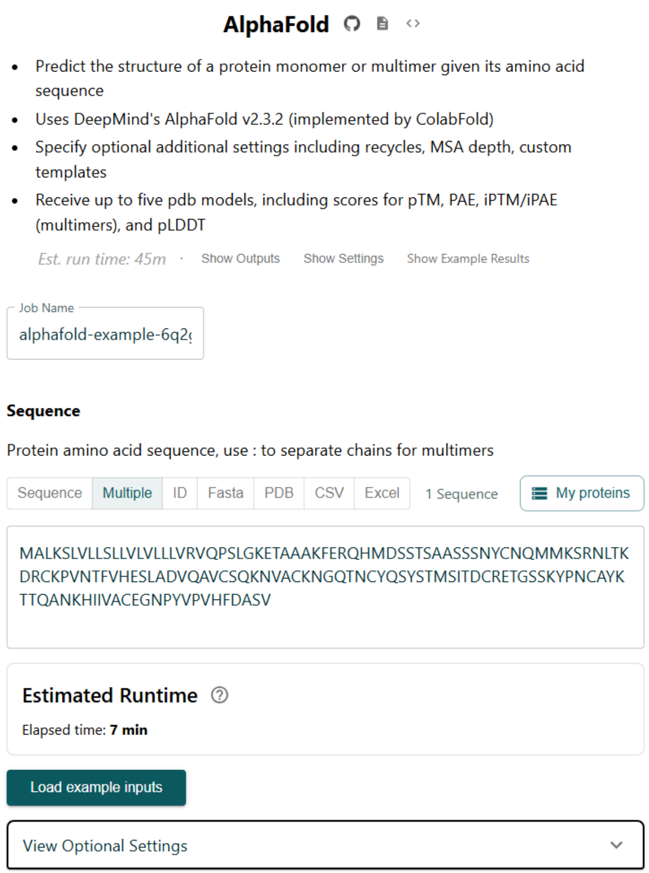
Why this is Interesting

This type of workflow is common in real research and industry platforms.

Instead of running models, scientists want to:

1. Run part of the model
2. Inspect intermediate signals
3. Modify parameters
4. Continue Pipeline

Making VizFold modular would support this type of exploration.



https://app.tamarind.bio/tools/alphafold